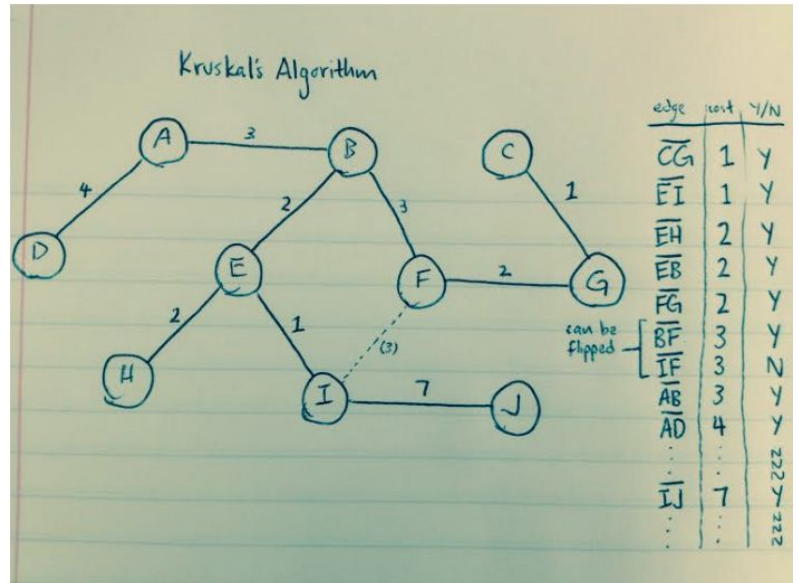


## 3134 Data Structures Homework 6 Written Solutions

1.(10 pts): Weiss, Exercise 9.15

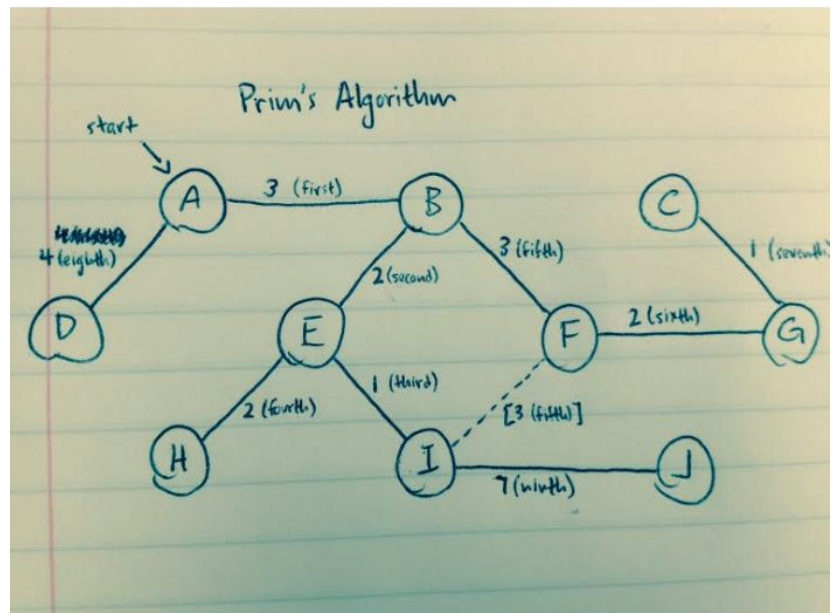
Kruskal's Algorithm

| edge  | cost | Y/N                       |
|-------|------|---------------------------|
| CG    | 1    | Y                         |
| EI    | 1    | Y                         |
| EH    | 2    | Y                         |
| EB    | 2    | Y                         |
| FG    | 2    | Y                         |
| BF IF | 3    | Y*can be one or the other |
| AB    | 3    | Y                         |
| AD    | 4    | Y                         |
| .     |      | N                         |
| .     |      | N                         |
| .     |      | N                         |
| IJ    | 7    | Y                         |
| .     |      | N                         |
| .     |      | N                         |
| .     |      | N                         |



Prim's Algorithm

| edge  | cost | order |
|-------|------|-------|
| AB    | 3    | 1     |
| BE    | 2    | 2     |
| EI    | 1    | 3     |
| EH    | 2    | 4     |
| BF IF | 3    | 5     |
| FG    | 2    | 6     |
| GC    | 1    | 7     |
| AD    | 4    | 8     |
| IJ    | 7    | 9     |



**Prim's, Book Version:**

1.

| <b>v</b> | <b>known</b> | <b>d<sub>v</sub></b> | <b>p<sub>v</sub></b> |
|----------|--------------|----------------------|----------------------|
| <b>A</b> | F            | 0                    | -                    |
| <b>B</b> | F            | ∞                    | -                    |
| <b>C</b> | F            | ∞                    | -                    |
| <b>D</b> | F            | ∞                    | -                    |
| <b>E</b> | F            | ∞                    | -                    |
| <b>F</b> | F            | ∞                    | -                    |
| <b>G</b> | F            | ∞                    | -                    |
| <b>H</b> | F            | ∞                    | -                    |
| <b>I</b> | F            | ∞                    | -                    |
| <b>J</b> | F            | ∞                    | -                    |

2.

| <b>v</b> | <b>known</b> | <b>d<sub>v</sub></b> | <b>p<sub>v</sub></b> |
|----------|--------------|----------------------|----------------------|
| <b>A</b> | T            | 0                    | -                    |
| <b>B</b> | F            | 3                    | A                    |
| <b>C</b> | F            | ∞                    | -                    |
| <b>D</b> | F            | 4                    | A                    |
| <b>E</b> | F            | 4                    | A                    |
| <b>F</b> | F            | ∞                    | -                    |
| <b>G</b> | F            | ∞                    | -                    |
| <b>H</b> | F            | ∞                    | -                    |
| <b>I</b> | F            | ∞                    | -                    |
| <b>J</b> | F            | ∞                    | -                    |

3.

| <b>v</b> | <b>known</b> | <b>d<sub>v</sub></b> | <b>p<sub>v</sub></b> |
|----------|--------------|----------------------|----------------------|
| <b>A</b> | T            | 0                    | -                    |
| <b>B</b> | T            | 3                    | A                    |
| <b>C</b> | F            | 10                   | B                    |
| <b>D</b> | F            | 4                    | A                    |
| <b>E</b> | F            | 2                    | B                    |
| <b>F</b> | F            | 3                    | B                    |
| <b>G</b> | F            | ∞                    | -                    |
| <b>H</b> | F            | ∞                    | -                    |
| <b>I</b> | F            | ∞                    | -                    |
| <b>J</b> | F            | ∞                    | -                    |

4.

| <b>v</b> | <b>known</b> | <b>d<sub>v</sub></b> | <b>p<sub>v</sub></b> |
|----------|--------------|----------------------|----------------------|
| <b>A</b> | T            | 0                    | -                    |
| <b>B</b> | T            | 3                    | A                    |
| <b>C</b> | F            | 10                   | B                    |
| <b>D</b> | F            | 4                    | A                    |
| <b>E</b> | T            | 2                    | B                    |
| <b>F</b> | F            | 3                    | B                    |
| <b>G</b> | F            | ∞                    | -                    |
| <b>H</b> | F            | 2                    | E                    |
| <b>I</b> | F            | 1                    | E                    |
| <b>J</b> | F            | ∞                    | -                    |

7.

| <b>v</b> | <b>known</b> | <b>d<sub>v</sub></b> | <b>p<sub>v</sub></b> |
|----------|--------------|----------------------|----------------------|
| <b>A</b> | T            | 0                    | -                    |
| <b>B</b> | T            | 3                    | A                    |
| <b>C</b> | F            | 6                    | F                    |
| <b>D</b> | F            | 4                    | A                    |
| <b>E</b> | T            | 2                    | B                    |
| <b>F</b> | T            | 3                    | B                    |
| <b>G</b> | F            | 2                    | F                    |
| <b>H</b> | T            | 2                    | E                    |
| <b>I</b> | T            | 1                    | E                    |
| <b>J</b> | F            | 7                    | I                    |

8.

| <b>v</b> | <b>known</b> | <b>d<sub>v</sub></b> | <b>p<sub>v</sub></b> |
|----------|--------------|----------------------|----------------------|
| <b>A</b> | T            | 0                    | -                    |
| <b>B</b> | T            | 3                    | A                    |
| <b>C</b> | F            | 1                    | G                    |
| <b>D</b> | F            | 4                    | A                    |
| <b>E</b> | T            | 2                    | B                    |
| <b>F</b> | T            | 3                    | B                    |
| <b>G</b> | T            | 2                    | F                    |
| <b>H</b> | T            | 2                    | E                    |
| <b>I</b> | T            | 1                    | E                    |
| <b>J</b> | F            | 7                    | I                    |

5.

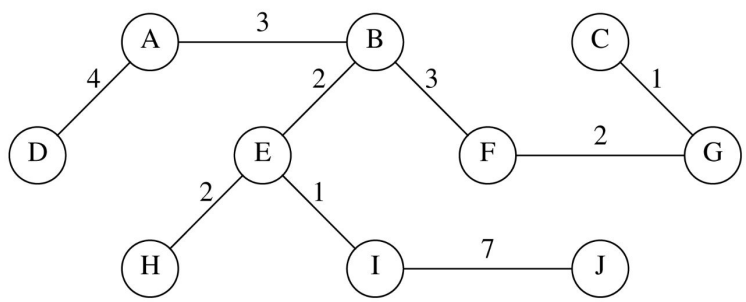
| <b>v</b> | <b>known</b> | <b>d<sub>v</sub></b> | <b>p<sub>v</sub></b> |
|----------|--------------|----------------------|----------------------|
| <b>A</b> | T            | 0                    | -                    |
| <b>B</b> | T            | 3                    | A                    |
| <b>C</b> | F            | 10                   | B                    |
| <b>D</b> | F            | 4                    | A                    |
| <b>E</b> | T            | 2                    | B                    |
| <b>F</b> | F            | 3                    | B                    |
| <b>G</b> | F            | $\infty$             | -                    |
| <b>H</b> | F            | 2                    | E                    |
| <b>I</b> | T            | 1                    | E                    |
| <b>J</b> | F            | 7                    | I                    |

6.

| <b>v</b> | <b>known</b> | <b>d<sub>v</sub></b> | <b>p<sub>v</sub></b> |
|----------|--------------|----------------------|----------------------|
| <b>A</b> | T            | 0                    | -                    |
| <b>B</b> | T            | 3                    | A                    |
| <b>C</b> | F            | 10                   | B                    |
| <b>D</b> | F            | 4                    | A                    |
| <b>E</b> | T            | 2                    | B                    |
| <b>F</b> | F            | 3                    | B                    |
| <b>G</b> | F            | $\infty$             | -                    |
| <b>H</b> | T            | 2                    | E                    |
| <b>I</b> | T            | 1                    | E                    |
| <b>J</b> | F            | 7                    | I                    |

\*FI could have been chosen instead of FB

Final MST



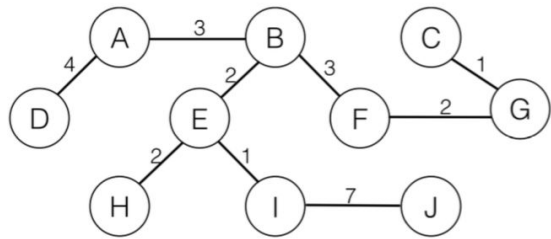
**Prim's, Class Version**

Expect to fill in a similar table on the final exam (either for Prim's, Dijkstra's, or unweighted shortest path).

| Step | Vertex Visited | A | B     | C        | D     | E     | F        | G        | H        | I        | J        |
|------|----------------|---|-------|----------|-------|-------|----------|----------|----------|----------|----------|
| 1    | A              | 0 | 3 (A) | $\infty$ | 4 (A) | 4 (A) | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 2    | B              | 0 | 3 (A) | 10 (B)   | 4 (A) | 2 (B) | 3 (B)    | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 3    | E              | 0 | 3 (A) | 10 (B)   | 4 (A) | 2 (B) | 3 (B)    | $\infty$ | 2 (E)    | 1 (E)    | $\infty$ |
| 4    | I              | 0 | 3 (A) | 10 (B)   | 4 (A) | 2 (B) | 3 (B)    | $\infty$ | 2 (E)    | 1 (E)    | 7 (I)    |
| 5    | H              | 0 | 3 (A) | 10 (B)   | 4 (A) | 2 (B) | 3 (B)    | $\infty$ | 2 (E)    | 1 (E)    | 7 (I)    |
| 6    | F              | 0 | 3 (A) | 6 (F)    | 4 (A) | 2 (B) | 3 (B)    | 2 (F)    | 2 (E)    | 1 (E)    | 7 (I)    |
| 7    | G              | 0 | 3 (A) | 1 (G)    | 4 (A) | 2 (B) | 3 (B)    | 2 (F)    | 2 (E)    | 1 (E)    | 7 (I)    |
| 8    | C              | 0 | 3 (A) | 1 (G)    | 4 (A) | 2 (B) | 3 (B)    | 2 (F)    | 2 (E)    | 1 (E)    | 7 (I)    |
| 9    | D              | 0 | 3 (A) | 1 (G)    | 4 (A) | 2 (B) | 3 (B)    | 2 (F)    | 2 (E)    | 1 (E)    | 7 (I)    |
| 10   | J              | 0 | 3 (A) | 1 (G)    | 4 (A) | 2 (B) | 3 (B)    | 2 (F)    | 2 (E)    | 1 (E)    | 7 (I)    |

Vertex names in ( ) indicate *prev* fields.  
Gray cells indicate settled/known vertices.

Final MST:



2.(10 pts): Weiss, Exercise 9.53 (From book solutions)

A: Each actor is a vertex; an edge connects two vertices if the actors have starred in the same movie. To find the Bacon number, simply find the shortest path between the given actor (source) and Kevin Bacon (target) using any of the algorithms covered in class.

B: Starting with Kevin Bacon, find all “paths” using DFS or another similar algorithm. Pick the longest path from the set of paths.

C: Similar to A, find minimal number of links between any two actors by setting one actor as source, another as target, and running any of the shortest-path algorithms.